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VHFACTMUDefTopic

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About the Time Measurement Unit

About the Time Measurement Unit

VHFACTime makes time measurements at the DUT. The VHFACTime instrument contains two asynchronous trigger lines.

This section includes the following topic:

- Features
- Safety Information

VHFACTime information includes the following sections:

- VHFAC TMU Theory of Operation
- VHFAC TMU Programming
- VHFAC Time Debug Display
- VHFAC TMU Examples

Related Topics

- VHFAC Specifications
- VHFACTime (VBT syntax)
- VHFAC DIB Design Guidelines

For information on other VHFAC instruments see:

- About the VHFAC Capture
- About the VHFAC Serial Bus
- About the VHFAC Source
- About the VHFAC Trigger Control

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About the Time Measurement Unit : Features

Features

VHFACTime can make the following time measurements:

- Frequency
- Period
- Pulse width
- Duty cycle
- Rise and fall time
- Pin-to-pin delay
- Jitter

VHFACTime allows asynchronous triggering of DUT pins.

VHFACTime has the following features:

- Slope select allows you to choose the rising or falling edge of a signal to start measurements.
- A front end with dual comparators

- Two time ranges:
  - Low range: 10 ms
  - High range: 20 s
- Dual time stamper
- Each stamper in a unit shares resources such as Start signals.
- Each stamper has its own dedicated memory of 512k capture samples.
- Each time stamper has three input signals:
  - event
  - enable
  - start
- Both stampers in the VHFACTime share the clock and start signals.
- Each stamper has its own event signal.
- Precounting allows signals to settle to a steady state before stamping.
  - Pre-Count
  - Between events
- Interleave mode

For VHFACTime hardware specifications, see VHFACTime.

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About the Time Measurement Unit : Safety Information

Safety Information

When operating the VHFACTime instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety for details.

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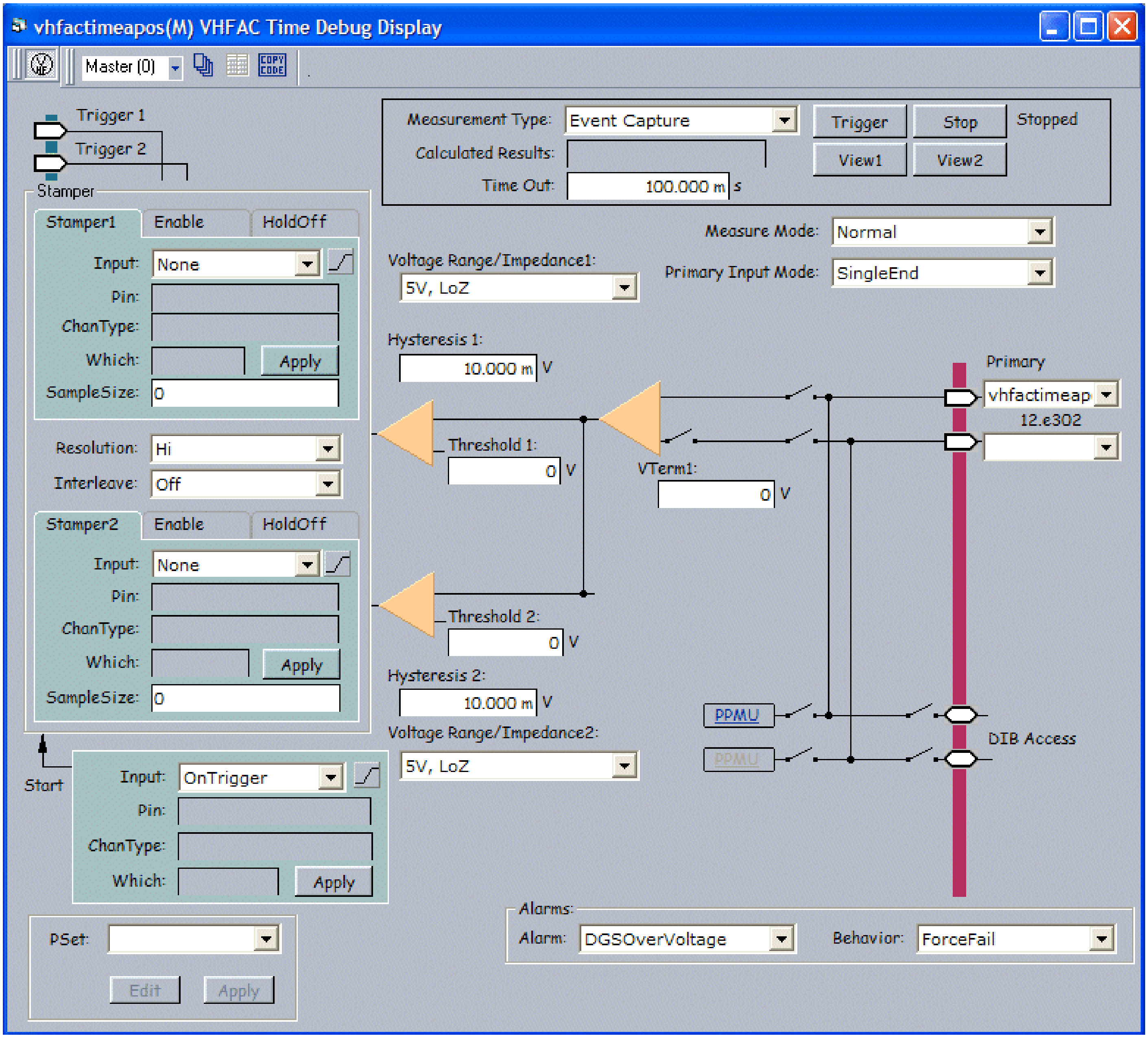
VHFAC Time Debug Display

VHFAC Time Debug Display

The VHFAC Time Debug Display is an interactive tool that shows the state of the VHFAC Time instrument in real time. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a drop-down menu. Non-interactive features are grayed out on the display and cannot be adjusted during debugging.



VHFAC Time Debug Display

This section describes features specific to the VHFAC Time debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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VHFAC Time Debug Display : Starting the VHFAC Time Debug Display

Starting the VHFAC Time Debug Display

To launch the VHFAC Time Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Time.

To start debugging, select a pin and a site to display the programmed values. Select sites and pins from the drop-down menu.

Without a test program loaded, the initial debug display returns an error:

! No pin exists of type VHFAC Time

and the connection lines appear directly connected to the DUT (without relays, DIB Access, or PPMU instruments active). Load a validated test program and restart the debug display to remedy this situation.

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VHFAC Time Debug Display : Debug Display Parameters and Buttons

Debug Display Parameters and Buttons

The VHFAC Time Debug Display contains several sections. Each section has programmable features used to make adjustments to test parameters.

The following topics are included:

- VHFAC Time Schematic
- VHFAC Time Alarm
- VHFAC Time Pin Setup Information

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VHFAC Time Debug Display : Debug Display Parameters and Buttons : VHFAC Time Schematic

VHFAC Time Schematic

The VHFAC Time Schematic section displays the current programmed settings of the selected VHFAC Time instrument and allows changes to be made to program parameters. Make connections to the DUT and other instruments by clicking the relay symbols.

The following topics are included:

- Stamper Tab
- Enable Tab
- HoldOff Tab
- Start Controls
- Additional Controls
- Front End Controls

Stamper Tab

VHFAC Time Schematic: Stamper Tab Parameters and Buttons

Parameter/Button	Description

Input	The event on which the time stamper becomes enabled. The following events can be set: • None • Pin • Pattern If Pin is selected, the Pin field is required. Otherwise, the Pin field is not required and is grayed out. (See Pin.)
Event Slope	Sets either negative or positive slope. The icon’s picture displays either a negative or positive slope. Slope does not apply when the stamper is off or when the Input is Pattern.
Pin	Selected from the programmed pins, this is the pin that enables the time stamper.
ChanType	This is the enable pin’s channel type. The enable channel type should match the channel type found in the specified channel map.
Which	Enter the value to identify trigger line. This is needed to identify trigger line that is used to asynchronously trigger the instrument.
SampleSize	Number of samples specified to capture the signal.

Enable Tab

VHFAC Time Schematic: Enable Tab Parameters and Buttons

Parameter/Button	Description
Input	The event on which the time stamper becomes enabled. The following events can be set: • None • Pin • Pattern

	If Pin is selected, the Pin field is required. Otherwise, the Pin field is not required and is grayed out. (See Pin.)
Enable Slope	Sets either negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is off or when the Input is Pattern.
Pin	Selected from the programmed pins, this is the pin that enables the time stamper.
ChanType	This is the enable pin's channel type. The enable channel type should match the channel type found in the specified channel map.
Which	Enter the value to identify trigger line. This is needed to identify trigger line that is used to asynchronously trigger the instrument.
Mode	The mode to determine how events are captured, whenever the enable input is another pin's signal. Select from the following: • Gate: Stamper captures events as long as its enable input is at gate_high. The stamper cannot stamp if enable input is gate_low. • TriggerEach: Stamper must receive a trigger on its enable input before capturing the next event signal.

HoldOff Tab

VHFAC Time Schematic: HoldOff Tab Parameters and Buttons

Parameter/Button	Description
Mode	The mode to specify which hold-off is used to allow time for a signal to settle before making a time measurement. The following modes can be set: • BeforeEach: Counts a programmed amount of time or counts a programmed number of events before making the first time stamp on each sample.

	• Off: No delay. • Precount: Counts a programmed amount of time or counts a programmed number of events before making the first time stamp on the first sample only.
Count	How many events to hold off before a signal is stamped. Hold off count waits for a certain number of events to occur before making a time measurement.

Start Controls

VHFAC Time Schematic: Start Parameters and Buttons

Parameter/Button	Description
Input	The event on which the start responds to. This following events can be set. • OnTrigger • Pattern • Pin The start is equivalent to and an on/off switch for the time stamper. If the stamper has received the start signal, then the stampers are on. Otherwise they are off.
Pin	Selected from the programmed pins, this is the pin of the Start signal.
ChanType	This is the start pin’s channel type. The start pin’s channel type should match the channel type found in the channel map that corresponds with the start pin.
Which	Enter 1 or 2. This identifies if threshold 1 or threshold 2 is used to asynchronously start the instrument.

	• 1: Threshold 1 • 2: Threshold 2
Slope	Sets either negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is off or when the Input is Pattern.

Additional Controls

VHFAC Time Schematic: Additional Parameters and Buttons

Parameter/Button	Description
Interleave	Select from the drop-down menu: • On: Stamper2 only captures events after Stamper1 has captured to avoid misalignments of the time stamps. • Off: Interleave mode is off and stamper can stamp in any order.
Resolution	Select from the drop-down menu: • Hi • Low Selects the range of resolution to stamp the time.
PSet	Lists the parameter sets (PSets) that where defined in language. To apply a PSet from the debug display, you must have previously defined PSets using VBT. Otherwise, the drop-down list is empty.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor in the TDE section.

Apply	Allows you to apply a PSet from the PSet list. See PSets in the VHFAC TMU Programming section.
Measurement Type	Measurement Type for the calculation (which will be indicated in the Calculated Results field). For example, if Period is set as Measurement Type, the calculation of period is displayed.
Calculated Results	Display the requested calculated result (calculation between stamper1 and stamper2 data).
Time Out	The programmed time out value.
Trigger	Initiates the instrument to start a capture.
View1	View the DSPWave for stamper1 (view the captured data on stamper1).
View2	View the DSPWave for stamper2 (view the captured data on stamper2).
Stop	Stop the measurement.
Measure Mode	Select from the drop-down menu: • Normal • SecondaryEnable • PinToPinDelay
Primary\Secondary Input Mode	Select from the drop-down menu: • SingleEnd • Differential

Front End Controls
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VHFAC Time Schematic: Front End Parameters and Buttons

Parameter/Button	Description

Threshold 1(2)	The value at which VHFAC Time should either stamp or trigger.
Hysteresis 1(2)	The hysteresis value to remove noise around the threshold crossing. • Hysteresis Maximum = 40 mV • Hysteresis Minimum = 1 mV
VTerm 1(2)	The value for negative input of the differential amplifier. • VTerm Maximum = -2.0 V • VTerm Minimum = 2.0 V
Voltage Range/ Impedance1(2)	Sets both voltage range and the impedance of the incoming signal so that it can be attenuated. Select a setting from the drop-down menu. Only a 5 V, LoZ mode is available.
PPMU	Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display.

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VHFAC Time Debug Display : Debug Display Parameters and Buttons : VHFAC Time Alarm

VHFAC Time Alarm

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during normal debug procedures:

VHFAC Time Alarm Parameters and Buttons

Parameter/Button	Description
Alarm	Select all alarms or the specific alarm to set an action for. Selecting an alarm only effects that alarm for that pin. • ADC1OutOfRange • ADC2OutOfRange • All • DGSOverVoltage • MeasurementIncomplete • StartEventNotArrived
Behavior	Choose an action for the alarm or alarms selected in Alarm. Alarm actions specify what should occur if the selected alarm condition

exists for the pin. This menu is grayed out until a selection is made in Alarm. Alarm behaviors are:

- Continue
- Default
- Off
- ForceBin
- ForceFail
- RaiseError

For more information, see VHFACTime Alarms.

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VHFAC Time Debug Display : Debug Display Parameters and Buttons : VHFAC Time Pin Setup Information

VHFAC Time Pin Setup Information

This information sets the pin and site to be debugged. Once a pin is selected, the debug display contains the pin’s current setup.

VHFAC Time Pin Setup Parameters and Buttons

Parameter/Button	Description
Pin (Primary and Secondary)	Lists the available pins from the pin map connected to the Time instrument. Selecting a pin from the list loads parameters for that pin to be debugged.
Site	This drop-down box, located on the debug display tool bar, lists the sites for the selected pin that are available. Selecting a site loads the parameters for the selected pin.
Apply All Sites	This icon is on the debug display tool bar. Clicking this button writes all changes made to every active site. If the button is not depressed, only the current active site is affected by the changes.

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VHFAC TMU Examples

VHFAC TMU Examples

This section provides basic examples and use cases for the VHFAC TMU.

The following topics are available:

- Perform a Frequency Measurement
- Perform a Period Measurement
- Perform a Rise Time Measurement
- Perform a Fall Time Measurement
- Perform a Duty Cycle Measurement
- Perform a Pulse Width Measurement
- Perform a Pin-to-Pin Delay (Propagation Delay) Measurement

For more information, see:

- VHFACTime (VBT Syntax Reference)
- VHFAC TMU Programming

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VHFAC TMU Examples : Perform a Frequency Measurement

Perform a Frequency Measurement

```
Public Function TMUFreqMeas(TMUPin As String, Threshold As Double, _
    HysteresisVal As Double, CapSize As Long, WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOff
        With .Measurement.Frequency
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            .Slope = tlSlopePositive
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
            .Threshold = Threshold
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.Frequency. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VHFAC TMU Examples : Perform a Period Measurement

Perform a Period Measurement

```
Public Function TMUPeriodMeas(TMUPin As String, Threshold As Double, _
    HysteresisVal As Double, CapSize As Long, WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.Period
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            .Slope = tlSlopePositive
            .Start.SetInput tlVHFACTimeStartInputPattern 'Trigger from pattern
            .Start.Slope = tlSlopePositive
            .Threshold = Threshold
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.Period. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VHFAC TMU Examples : Perform a Rise Time Measurement

Perform a Rise Time Measurement

```
Public Function TMURiseTimeMeas(TMUPin As String, Threshold1 As Double, _
    Threshold2 As Double, HysteresisVal As Double, CapSize As Long, _
    WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.RiseTime
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            Call .SetThresholds(Threshold1, Threshold2)
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.RiseTime. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VhfacTMU_example.5.5

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VHFAC TMU Examples : Perform a Fall Time Measurement

Perform a Fall Time Measurement

```
Public Function TMUFallTimeMeas(TMUPin As String, Threshold1 As Double, _
    Threshold2 As Double, HysteresisVal As Double, CapSize As Long, _
    WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.FallTime
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            Call .SetThresholds(Threshold1, Threshold2)
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.FallTime. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VhfacTMU_example.5.6

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VHFAC TMU Examples : Perform a Duty Cycle Measurement

Perform a Duty Cycle Measurement

```
Public Function TMUDutyCycleMeas(TMUPin As String, Threshold As Double, _
    HysteresisVal As Double, CapSize As Long, WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.DutyCycle
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            .Slope = tlSlopePositive
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
            .Threshold = Threshold
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 Measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.DutyCycle. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VhfacTMU_example.5.7

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VHFAC TMU Examples : Perform a Pulse Width Measurement

Perform a Pulse Width Measurement

```
Public Function TMUPulseWidthMeas(TMUPin As String, Threshold As Double, _
    HysteresisVal As Double, CapSize As Long, WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(TMUPin)
        .Resolution = tlVHFACTimeResolutionHi
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.PulseWidth
            .Hysteresis = HysteresisVal
            .SampleSize = CapSize
            .SetInput TMUPin, "VHFACTimePrimaryPos", 1, 1
            .Slope = tlSlopePositive
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
            .Threshold = Threshold
```

```
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(TMUPin).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 Measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(TMUPin).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(TMUPin).Measurement.PulseWidth. _
        CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(TMUPin).Disconnect

End Function
```

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VHFAC TMU Examples : Perform a Pin-to-Pin Delay (Propagation Delay) Measurement

Perform a Pin-to-Pin Delay (Propagation Delay) Measurement

```
Public Function TMUPinToPinDelayMeas(VHFACTimePinA As String, _
    VHFACTimePinB As String,Threshold1 As Double, Threshold2 As Double, _
    HysteresisVal As Double, CapSize As Long, WaitTime As Double) As Long

    Dim stm1 As New PinListData
    Dim stm2 As New PinListData
    Dim ResultDSPWave As New DSPWave
    Dim MeanResult As New SiteDouble
    Dim Site As Variant

    ' Connect VHFAC TMU A and B input pins
    TheHdw.VHFACTime.Pins(VHFACTimePinA).Connect
    TheHdw.VHFACTime.Pins(VHFACTimePinB).Connect

    ' Set up VHFAC TMU parameters
    With TheHdw.VHFACTime.Pins(VHFACTimePinA)
        .Resolution = tlVHFACTimeResolutionHi
        .MeasureMode = tlVHFACTimeMeasureModePinToPinDelay
        .Interleave = tlVHFACTimeInterleaveOn
        With .Measurement.PinToPinDelay
            .SampleSize = CapSize
            Call .SetHysteresis(HysteresisVal, HysteresisVal)
            .SetInput1 VHFACTimePinA, "VHFACTimePrimaryPos", 1, 1
            .SetInput2 VHFACTimePinB, "VHFACTimeSecondaryPos", 2, 1
```

```
            Call .SetSlopes(tlSlopePositive, tlSlopePositive)
            Call .SetThresholds(Threshold1, Threshold2)
            .Start.SetInput tlVHFACTimeStartInputOnTrigger
            .Start.Slope = tlSlopePositive
        End With
    End With

    ' Trigger the VHFAC TMU Capture
    With TheHdw.VHFACTime.Pins(VHFACTimePinA).TimeStamps
        .Trigger ("Capture")
        .TimeOut (WaitTime)
    End With

    ' Get Stamper1 and Stamper2 Measurements and perform calculation to get
    ' results
    TheHdw.VHFACTime.Pins(VHFACTimePinA).TimeStamps.Item("Capture"). _
        GetWaves stm1, stm2
    ResultDSPWave = TheHdw.VHFACTime.Pins(VHFACTimePinA).Measurement. _
        PinToPinDelay.CalculatedResults(stm1, stm2)

    ' In a site loop, calculate the mean value for each site; return results
    ' into SiteDouble
    For Each Site In TheExec.Sites
        MeanResult = ResultDSPWave.CalcMean
    Next Site

    ' Disconnect VHFAC TMU
    TheHdw.VHFACTime.Pins(VHFACTimePinA).Disconnect
    TheHdw.VHFACTime.Pins(VHFACTimePinB).Disconnect

End Function
```

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VHFAC TMU Programming

VHFAC TMU Programming

This section describes programming of a VHFAC Time Measurement Unit.

The following topics are available:

- VHFAC TMU DataTool Setup
- VHFAC TMU Pattern Control

For more information, see:

- VHFACTime (VBT Syntax Reference)
- VHFAC TMU Examples

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VHFAC TMU Programming : VHFAC TMU DataTool Setup

VHFAC TMU DataTool Setup

This section describes how to program several DataTool worksheets to handle basic VHFAC TMU parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map

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VHFAC TMU Programming : VHFAC TMU DataTool Setup : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. For the VHFAC TMU, the connections that you must enter in the pin map are analog channel connections to the DUT.

You can enter any pin type for the VHFAC instrument. The most common selection for VHFAC instruments is:

Analog	This type is the primary type used for any of the VHFAC instruments.

For more information, see Pin Map Sheet in the DataTool section.

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VHFAC TMU Programming : VHFAC TMU DataTool Setup : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the VHFAC TMU instrument, you must select a channel type from the Type column menu for each pin or pin group. Valid VHFAC TMU channel types are:

VHFACTimePrimaryPos	Primary positive
VHFACTimePrimaryNeg	Primary negative
VHFACTimeSecondaryPos	Secondary positive
VHFACTimeSecondaryNeg	Secondary negative

Type the tester channel information value for each site to connect a DUT pin to the instrument.This table lists the available VHFAC Time Measurement channels. Note that in the table, x indicates the DIB slot number of the VHFAC TMU.

VHFAC Time Measurement Unit Channel Resources

VHFAC Capture Pogo Pin	DIB Channel	Signal Name
VHFAC TMU Primary Pos	x.a27	x.tmuapos
VHFACTMU Primary Neg	x.b26	x.tmuaneg
VHFACTMU Secondary Pos	x.a29	x.tmubpos
VHFAC TMU Secondary Neg	x.b28	x.tmubneg

For more information, see Channel Map Sheet in the DataTool section.

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VHFAC TMU Programming : VHFAC TMU Pattern Control

VHFAC TMU Pattern Control

The VHFAC instruments can be controlled by patterns and microcode allowing changes to instrument states and parameters during a program. A very accurate level of control can be obtained by programming an instrument with micrcodes, patterns and PSets.

A pattern is created independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling pattern execution.

The two forms of pattern control are:

Microcode	A microcode controls a single function, such as gating an instrument on or off or starting a waveform segment for example.
PSets	A PSet defines a programmed state for an instrument. When an instrument receives a PSet from a pattern, it enters the programmed state represented by that PSet.

The following topics are available:

- Patterns
- VHFAC TMU Microcodes
- PSets

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VHFAC TMU Programming : VHFAC TMU Pattern Control : Patterns

Patterns

A pattern is used to precisely program an instrument at the vector level. Parameters can be programmed using microcode or by calling PSets. The following tools are used in pattern creation.

Pattern Compiler

The pattern compiler takes a text .atp file and compiles it into a binary .pat file. This file can be edited using the PatternTool and is the format accepted by the pattern loader.

Pattern Loader

The pattern loader takes the binary .pat file and loads it onto the hardware, along with the loading instruction memory, target memory, and waveform memory.

PatternTool

PatternTool allows you to read and edit binary pattern files. Using PatternTool, you can display a compiled pattern file in easily readable, column formats. You can also directly edit the pattern file, instead of having to edit the ASCII source file and recompile it. You can then update the tester with the edited pattern file.

PatternTool is useful during debugging, specifically in debugging pattern files. PatternTool can also display HRAM (History RAM) data, as captured on the tester.

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VHFAC TMU Programming : VHFAC TMU Pattern Control : VHFAC TMU Microcodes

VHFAC TMU Microcodes

A microcode controls a single function such as gating on or off. The table lists VHFACTime microcodes.

VHFACTime Microcodes

Microcode	Description
Enable_Trig	Sets the instrument to respond to a trigger signal.
Trig <name>	Starts a capture with the given name as specified in the most recent PSet.
Stamp	Causes a time stamp on the time stamper set up with a pattern as the event input.
Enable_Gate_On	Turns on the Enable Gate window. While this Gate is open, the time stamper is enabled.
Enable_Gate_Off	Turns off the Enable Gate window.
PSet <name>	Applies the PSet of the given name.
Enable_Alarm	Open alarm window for VHFACTime alarms.
Disable_Alarm	Close alarm window for VHFACTime alarms.

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VHFAC TMU Programming : VHFAC TMU Pattern Control : PSets

PSets

PSets control the state of an instrument for testing, such as the force voltage value and force voltage range. PSets are programmable, and the characteristics are set up using its parameters. You define a PSet in VBT language and invoke it using PSet microcode. The list of PSet properties and methods for VHFACTime are listed in the table.

PSets allow you to specify several aspects of an instrument’s state at once.

VHFACTime PSets

VHFACTime PSet Property	Parameter	Value (Min, Max)	Units
FrontEnd	Specifies the front end properties.		
	Hysteresis1	(0.001, 0.05)	V
	Hysteresis2	(0.001, 0.05)	V
	Threshold1	(-5, 5)	V
	Threshold2	(-5, 5)	V
	VTerm1	(-2, 2)	V
	VTerm2	(-2, 2)	V
Interleave	Sets interleave mode on or off.		
	Interleave	On, Off	
Resolution	Sets resolution high or low		
	Resolution	Hi, Lo	

Stamper1	Specifies stamper1 properties.	
	Enable Input	None, Pin(Time's pin, others pin), Pattern
	Enable Mode	Gate, TriggerEach
	Enable Slope	Pos, Neg
	Event Input	None, Pin(Time's pin, others pin), Pattern
	Event Slope	Pos, Neg
	Holdoff Count	(0, 4095)
	Holdoff Mode	Off, Precount, BeforeEach
	Sample Size	(0, 524287)
Stamper2	Specifies stamper2 properties	
	Enable Input	None, Pin(Time's pin, others pin), Pattern
	Enable Mode	Gate, TriggerEach
	Enable Slope	Pos, Neg
	Event Input	None, Pin(Time's pin, others pin), Pattern
	Event Slope	Pos, Neg
	Holdoff Count	(0, 4095)
	Holdoff Mode	Off, Precount, BeforeEach
	Sample Size	(0, 524287)
Start	Species the start interface including slope information.	
	Input	OnTrigger, Pin(Time's pin, others pin), Pattern
	Slope	Pos, Neg

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VHFAC TMU Theory of Operation

VHFAC TMU Theory of Operation

The VHFAC TMU is a dual time stamper time measurement instrument with programmable front-end threshold, hysteresis and slope, two stamp resolution modes, enable input and asynchronous trigger capabilities, as well as other useful time measurement features.

The following topics are available:

For information on VHFAC TMU instrument performance specifications and standards, see VHFAC Specifications.

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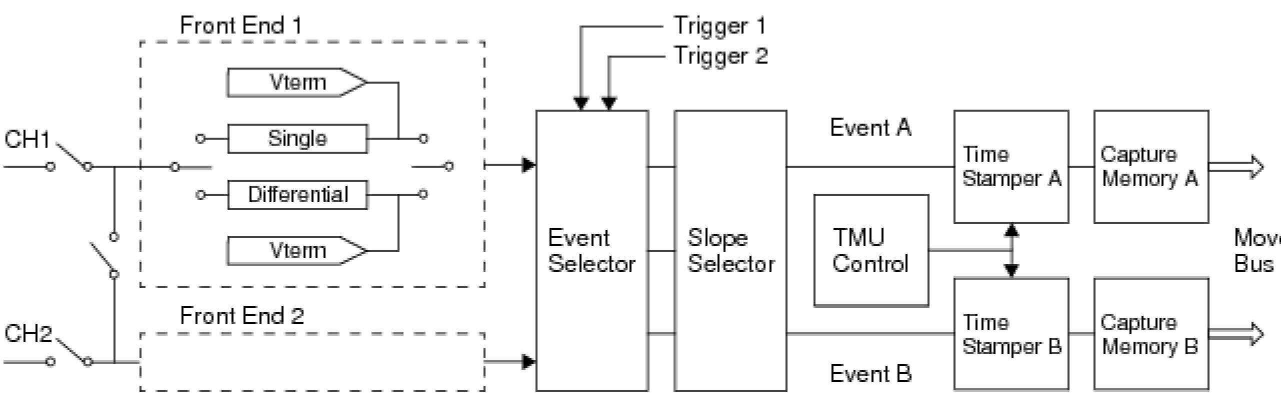
VHFAC TMU Theory of Operation : Overview

Overview

A unique feature of the VHFAC TMU is the existence of two independent front ends as well as two differential input paths. The primary purpose of these features is to perform accurate pin-to-pin delay (propagation delay) measurements with a single VHFAC TMU instrument.

VHFACTime Channel Access

VHFACTime	DC30 Channels	DC75 Channels
1	1, 2, 5, 6, 9, 10, 13, 14, 17, 18	1, 2
2	3, 4, 7, 8, 11, 12, 15, 16, 19, 20	3, 4



VHFAC Time Measurement Unit Block Diagram

The two differential input paths are denoted above as CH1 and CH2 (also referred to as PinA and PinB in the diagrams below), referring to the primary and secondary inputs respectively. IG-XL allows you to program the instrument in three measure modes. See Measure Modes.

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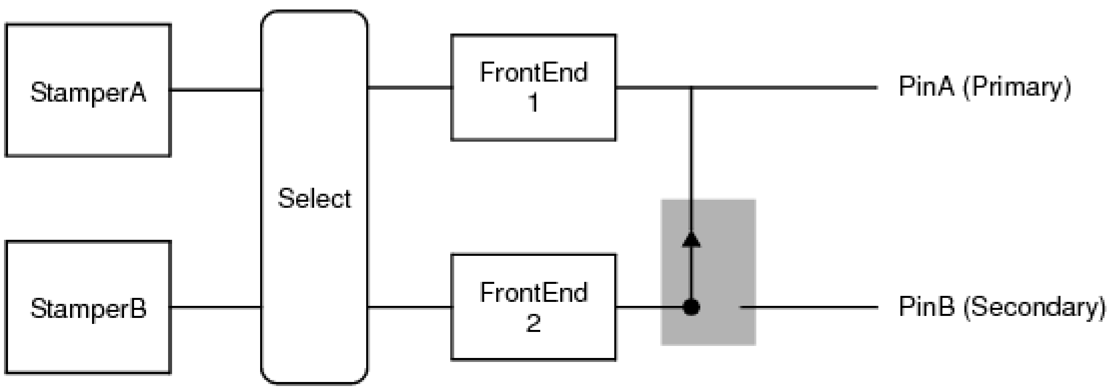
VHFAC TMU Theory of Operation : Measure Modes

Measure Modes

Normal Mode

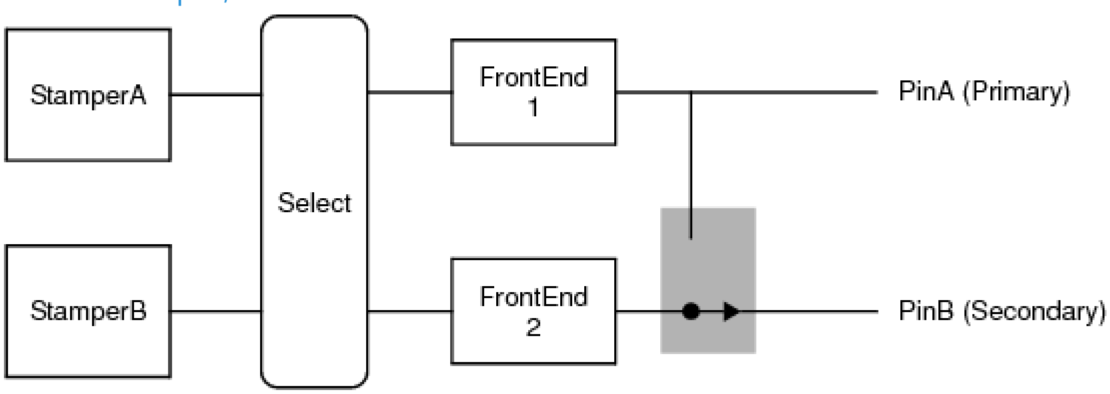
Normal mode is used to perform all basic time measurements with event signals entering the instrument through the primary input path. In Normal mode the secondary input path is inactive, and therefore cannot be used.



Normal Measure Mode

Secondary Enable Mode

Secondary Enable mode is used to perform all basic time measurements with event signals entering the instrument through the primary input path. In this mode, the secondary input path is used as an enable input, and cannot be used to measure events.

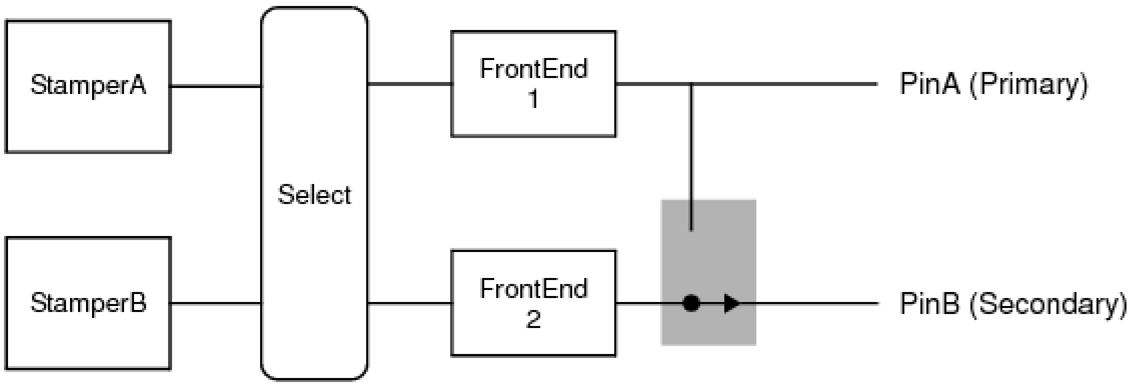


Secondary Enable Measure Mode

Pin-to-Pin Delay Mode

The primary purpose of Pin-to-Pin Delay Mode is to measure the time between events on different pins (such as Propagation Delay). In this mode both the primary and secondary input paths are configured as event inputs.

Alternatively, through proper programming of the instrument, the secondary input path can receive event signals, independent of the primary input path, to perform a subset of basic time measurements (that is, perform time measurements with secondary input pins). See Performing Basic Time Measurements with the Secondary Input Path for details.



Pin-to-Pin Delay Measure Mode

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VHFAC TMU Theory of Operation : Measure Modes : Performing Basic Time Measurements with the Secondary Input Path

Performing Basic Time Measurements with the Secondary Input Path

VHFAC TMU hardware is designed such that the primary input path can gain access to both FrontEnd 1 and FrontEnd 2. However, the secondary input path can only gain access to FrontEnd 2. Normally when programming the instrument to perform basic time measurements, the time stampers expect events to pass through either a combination of FrontEnd 1 and FrontEnd 2 (rise time and fall time measurements) or FrontEnd 1 only (frequency, period, pulse width, and duty cycle measurements).

In the case of events entering the instrument through the secondary path, the hardware must be configured such that the time stampers will expect events through FrontEnd 2, since the hardware does not allow access to FrontEnd1 from the secondary path. Because the secondary input has access to only one front end, only measurements requiring one front end can be performed. Specifically, this means measurements such as rise time and fall time cannot be performed with the secondary input. However, measurements such as frequency, period, pulse width, and duty cycle can be performed with the secondary input.

To configure the hardware in this manner, follow the steps below:

1. Connect the secondary DUT pin to the DIB.
2. Program the measure mode to be PinToPinDelay.
3. Program the stamper event input to connect to FrontEnd 2.
4. Program the remaining instrument parameters for specific time measurement.

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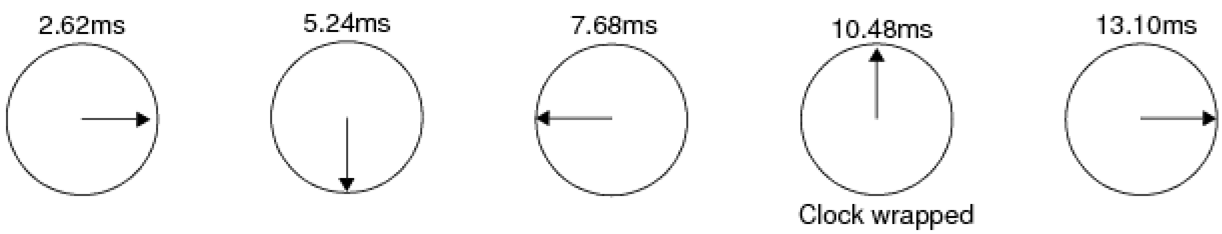
VHFAC TMU Theory of Operation : VHFACTime Clock

VHFACTime Clock

The VHFACTime has one clock for the dual stamper, which means two stampers share one clock. The VHFACTime clock is a circular clock. A circular clock is a continuos running clock that wraps over at a certain point rather than maintaining a running total of time. One full rotation through the VHFACTime clock is10.48 ms for the low range and 21.47s for the high range. On reaching this point, the clock resets to zero and repeats the rotation. The hardware and software does track if the clock has wrapped. However, you cannot use the clock exceeding its time range. When a first sample is captured, its clock value is stored in the hardware. The programmed number of samples must finish the capture within one cycle after the first sample is captured. If some samples are not finished in one cycle and the clock reaches the first stored value, VHFACTime created an alarm and resets the hardware. The figure shows the clock wrapping through time.



VHFACTime Clock Wrapping Example

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VHFAC TMU Theory of Operation : Measurement Types

Measurement Types

The VHFACTime can make the following types of measurements:

- Period
- Frequency
- Pulse width
- Duty cycle
- Pin-to-pin delay
- Rise or fall times
- Jitter

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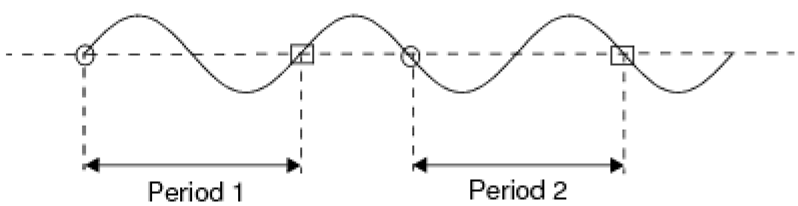
VHFAC TMU Theory of Operation : Measurement Types : Period

Period

Computes the time from the capture of an event on the input signal until a second capture of the same event on the same signal. In the figure stamper 1 stamps the beginning and stamper 2 stamps the finish. The time between the beginning and finish is the period. The finish stamp for one period cannot be used as the start stamp for a second period measurement. A new start stamp must occur.



Example of a Period Measurement

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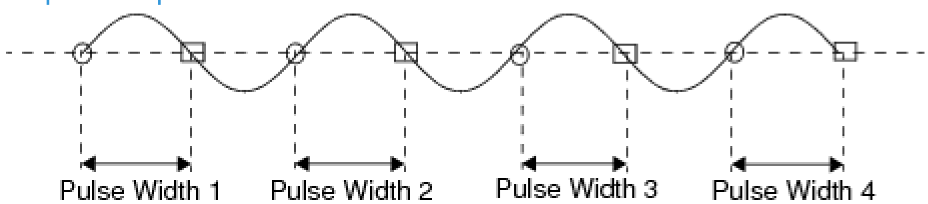
VHFAC TMU Theory of Operation : Measurement Types : Pulse Width

Pulse Width

Computes the time from an event on the input signal to another event on the input signal with the identical threshold voltage and the opposite slope. This figure shows stamper 1 stamping the beginning on the rising (positive) slope and stamper 2 stamping the finish on the falling (negative) slope. The pulse width is the difference between the two.



Example of Pulse Width Measurement

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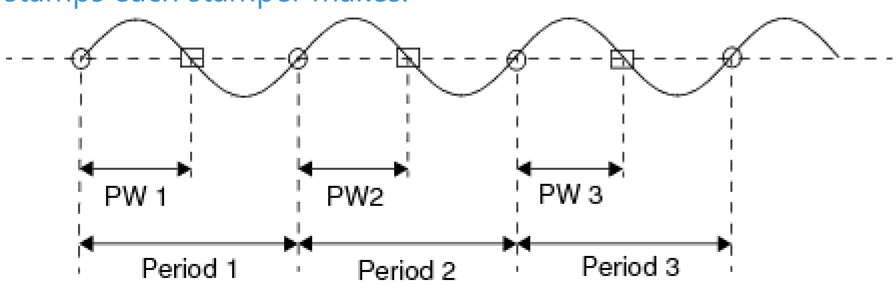
VHFAC TMU Theory of Operation : Measurement Types : Duty Cycle

Duty Cycle

Computes the ratio of time in one period that the input signal was above or below a specified threshold. To calculate duty cycle both a pulse width and a period must be measured. For the pulse width stamper 1 stamps the beginning and stamper 2 stamps the finish. For the period measurement stamper 1 stamps both the beginning and finish. This figure shows an example of the stamps each stamper makes.



Example of Duty Cycle Measurement

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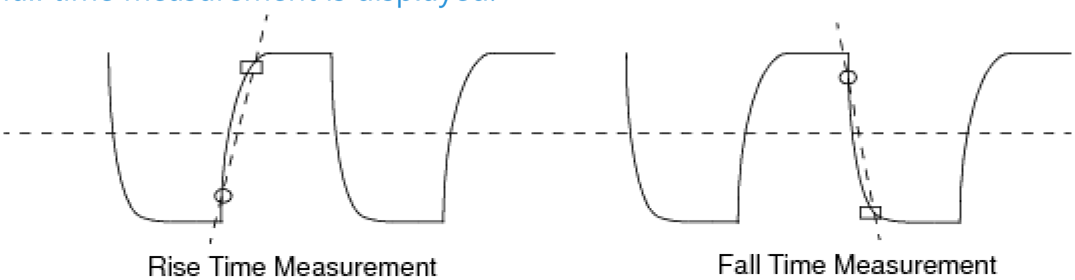
VHFAC TMU Theory of Operation : Measurement Types : Rise or Fall Time

Rise or Fall Time

Computes the time an input signal takes to travel between two thresholds with the same slope. Figure shows stamper 1 stamps the beginning and stamper two stamps the finish. Both a rise and fall time measurement is displayed.



Example of Rise and Fall Time Measurement

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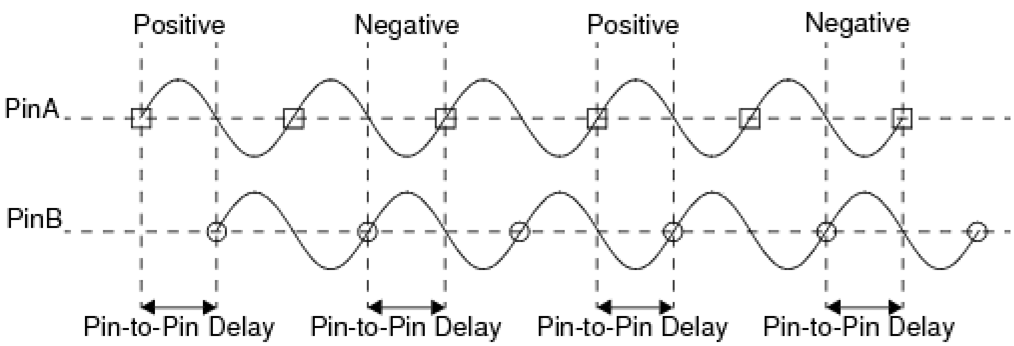
VHFAC TMU Theory of Operation : Measurement Types : Pin-to-Pin Delay

Pin-to-Pin Delay

Computes the time between a specified event on one input and a separately specified event on another input. Each input is connected to a separate VHFACTime Stamper on the same VHFAC instrument board. The figure shows Pin A connected to Stamper 1 of a VHFACTime and Pin B connected to Stamper 2 of the VHFACTime on the same VHFAC board. The difference between the stamp on Pin A and stamp on Pin B is the pin-to-pin delay.



Example of Pin-to-Pin Delay Measurement

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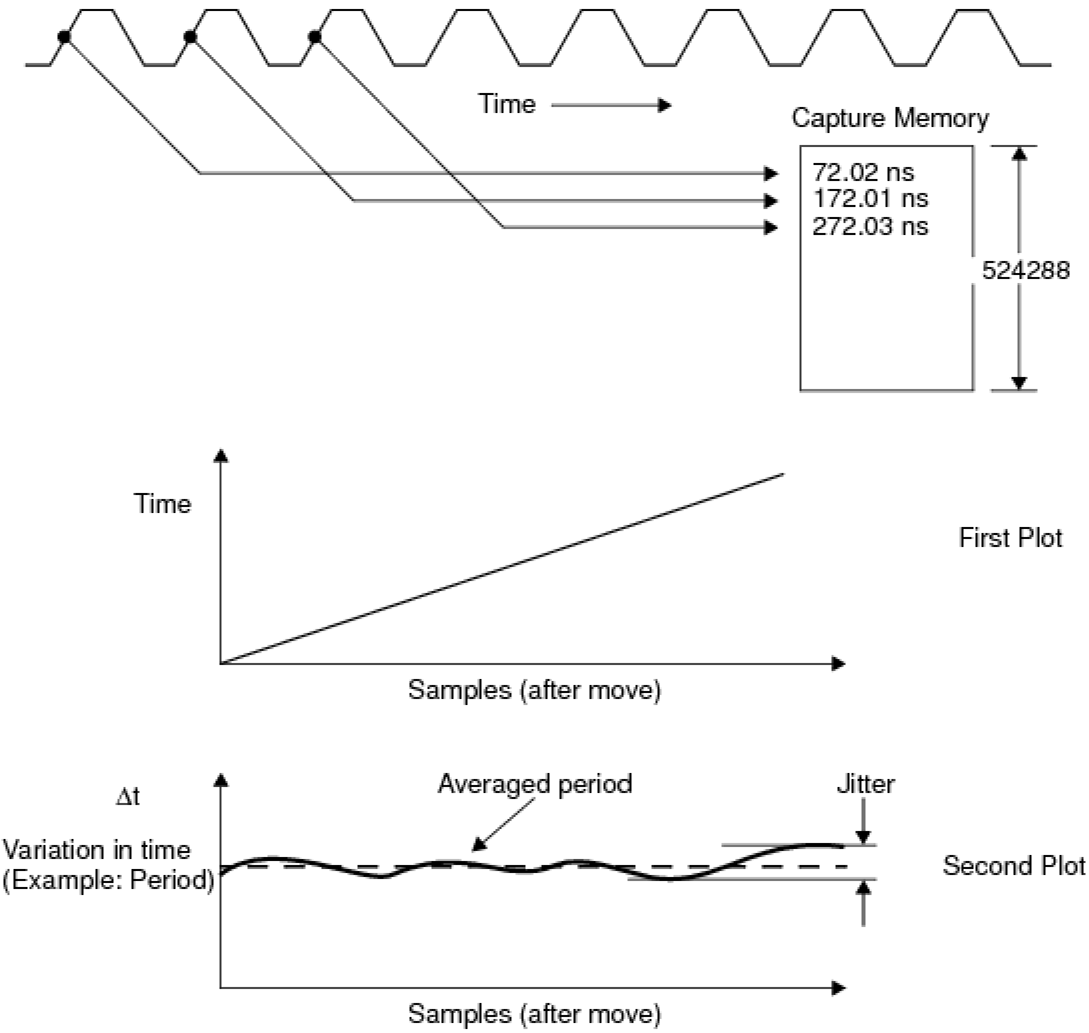
VHFAC TMU Theory of Operation : Measurement Types : Jitter

Jitter

VHFACTime is implemented with a time stamping architecture. There is a fundamental difference between time stamping and time measurement. A time measurement starts and stops a timer, while a time stamper records or stamps the time of each event. A functional block of the VHFACTime hardware attaches a time stamper to capture and record the occurrence of an event. In this example, the figure shows a DUT signal with detected events. The time of each event is stored in memory. Also shown are two plots. The first plot shows captured event time versus sample number. For events evenly spaced in time (a periodic signal), the time value should increase linearly with the sample number.



Theory of Jitter Measurement

The second plot is an example of the variation in time that can be derived from the captured event time. The plot shows the variation in the period of the signal. The average period and the period jitter can be calculated.

Since the captured data consists of the times of occurrence of specified events, you can determine the parameter of interest after data is collected. For example, consider measuring the alignment of random data transitions to a clock edge. Using time stamping, we can define three events: data rising, data falling, and clock. Each of these events in the data stream is stamped, and the captured data can be analyzed for the parameters of interest.

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VHFAC TMU Theory of Operation : Start Modes

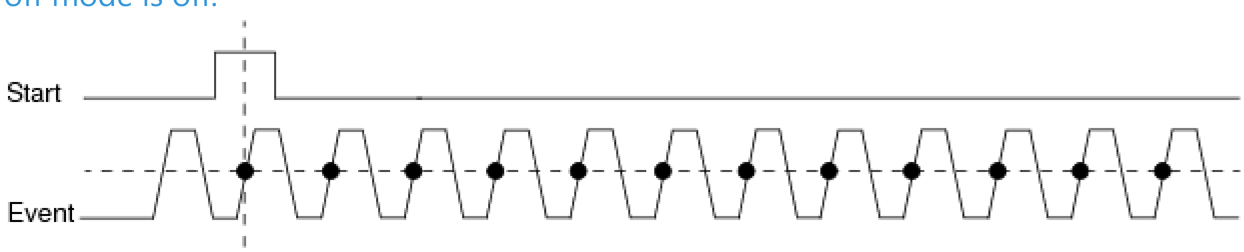
Start Modes

When a start signal is received, the VHFACTime makes measurements until it finishes the programmed capture or until a stop command is issued. The start signal is made only for a single specified measurement or for all programmed measurements. When the start is executed, the program determines which instrument is being used to make the measurement, checks the programmed settings and starts the measurement.

The VHFACTime has three start modes:

Start	The measurement automatically begins with the first event detected after the start command is generated.
External	The Time stamper can be started by a command over the Trigger Matrix.
Microcode	The start command can be generated in microcode

VHFACTime has one start signal which is shared by both stampers. The measurement begins on the next event after the start event. The figure shows a start signal from an external event and captures on each event after the start. This example assumes enable input is programmed to none and hold off mode is off.



Example of Starting the Instrument in Continuous Mode Using an External Event

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VHFAC TMU Theory of Operation : Enable Mode

Enable Mode

Enable allows you to control which edges are captured. After the start signal the stamper is ready to stamp and is just waiting for an enable signal.

Enable Mode controls when each individual stamper can stamp. VHFACTime has three modes of selectively capturing events:

Continuous (off)	Once started, the stamper captures all subsequent events.
Trigger Each Event	Stamper must receive a trigger on its enable input before capturing the next event signal.
Gate Window	Stamper captures events as long as its enable input is at gate_high. The stamper cannot stamp if enable input is gate_low.

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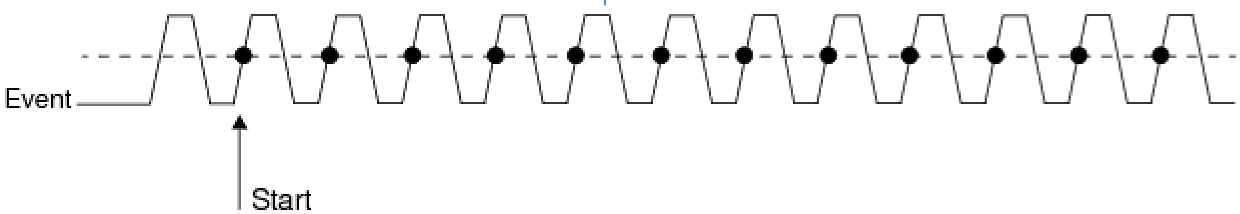
VHFAC TMU Theory of Operation : Enable Mode : Continuous Mode

Continuous Mode

This figure shows the edges captured on a signal when the enable mode is programmed to Continuous. Each event that occurs is stamped.



Continuous Mode

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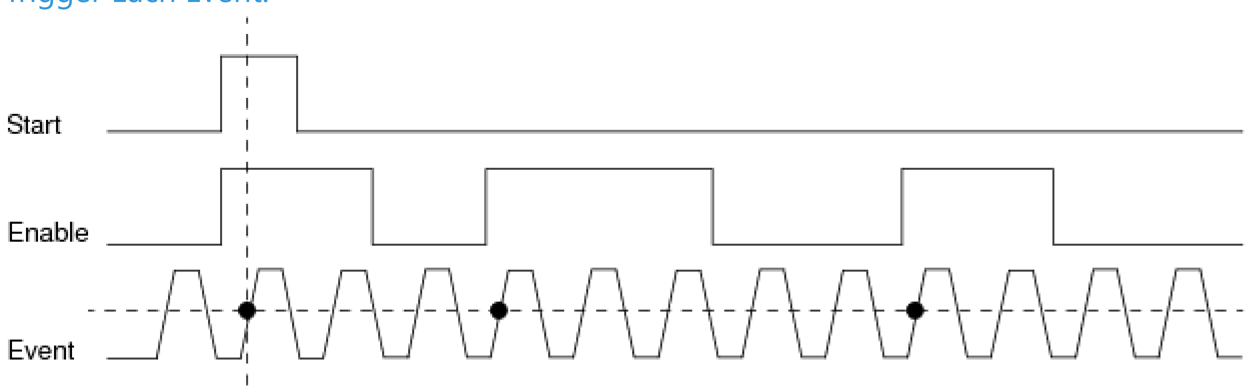
VHFAC TMU Theory of Operation : Enable Mode : Trigger Each Event

Trigger Each Event

Trigger Each enables the capture of a single event on the enable signal's set edge. The edge choice can be set by positive or negative slope.

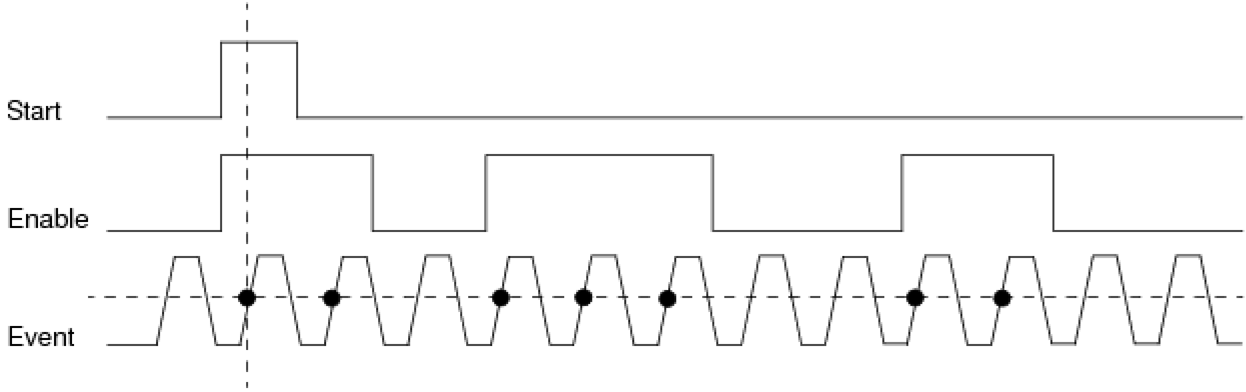
Trigger mode enables the capture of an event or events during testing. Once the instrument is started, the first event after each trigger can be captured. Triggering allows you to control which edges are captured. After the start signal the stamper is ready to stamp, just waiting for an enable signal. The figure shows the enable signal and the event that is stamped. Even though the enable signal is active for more than one event, only the first event after the enable is captured using Trigger Each Event.



Trigger Each Event Mode

Gate Window

Gating enables the capture of one or more events by setting the level of a gate signal. After the start signal the stamper is ready to stamp, just waiting for an enable signal. The figure shows the enable signal and the events stamped. While the stamper is enabled in Gate Window each event that occurs is stamped.



Example of Gating the Instrument With Enable On

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VHFAC TMU Theory of Operation : Enable Mode : Event Signal

Event Signal

The event is the signal that the VHFACTime records time on. It is this signal that the stampers stamp a time on. The event signal has six settings:

None	The event signal is off and not used.
Pattern	The event signal come from a pattern.
Threshold 1	The event occurs when the programmed threshold 1 is reached.
Threshold 2	The event occurs when the programmed threshold 2 is reached.
Trigger1	The event occurs when Trigger1 is used.
Trigger2	The event occurs when Trigger2 is used

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VHFAC TMU Theory of Operation : Enable Mode : Hold-Off Mode

Hold-Off Mode

Hold-off mode is used to create a delay before a stamp is made to allow a hold-off to account for signal settling. Hold-off can be set to the following:

Precount once	Counts a programmed amount of time or counts a programmed number of events before taking each time stamp.
Precount between events	Counts a programmed amount of time or counts a programmed number of events before making the first time stamp on each sample.
Off	No delay.

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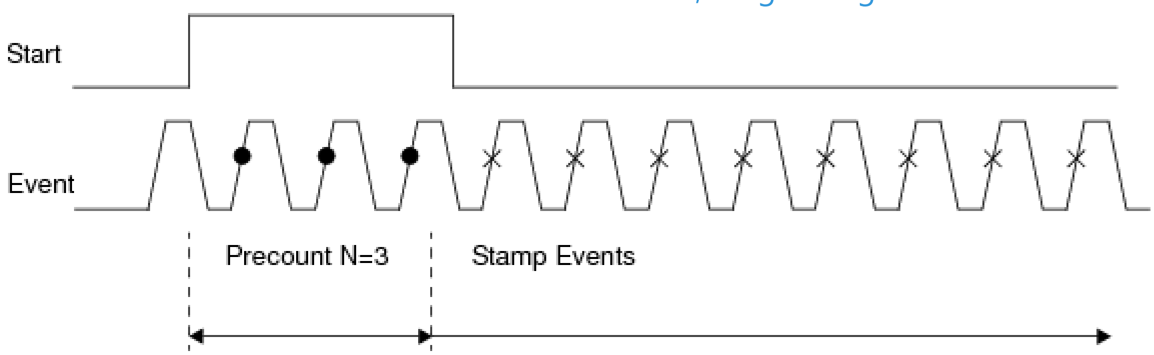
VHFAC TMU Theory of Operation : Enable Mode : Precount Once

Precount Once

When the Hold-off mode is set to Precount Once, a signal is given time to settle before stamping:



Precount Once

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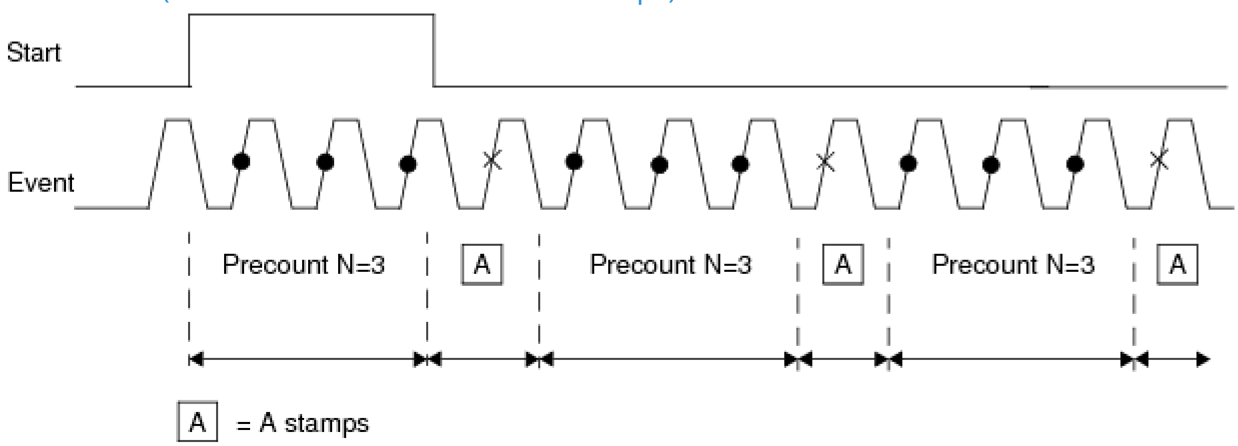
VHFAC TMU Theory of Operation : Enable Mode : Precount Between Events

Precount Between Events

When the Hold-off mode is set to Precount Between Events (that is, precount before each event), more samples are taken to achieve greater accuracy in the frequency measurements. Also, this feature increases the maximum measurable frequency. The duration of stamps must be greater than 80 ns (the minimum time between stamps).



Precount Between Events

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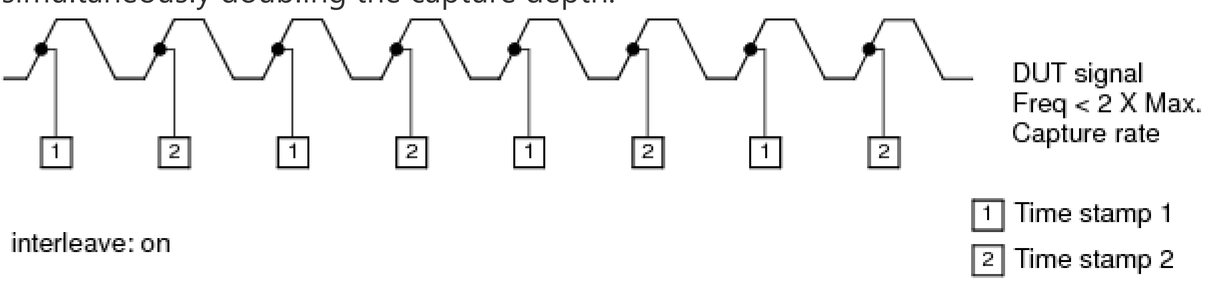
VHFAC TMU Theory of Operation : Interleave Mode

Interleave Mode

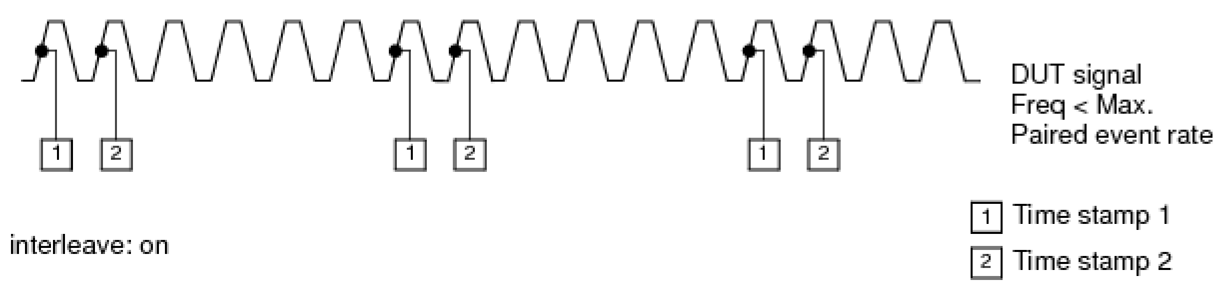
By interleaving the stampers, the maximum capture frequency doubles. In interleave mode, Time stamper 1 and stamper 2 alternate event detection and stamping. This allows the detection and stamping rate of events to be twice the maximum stamping rate of a single time stamper while simultaneously doubling the capture depth.



Interleave Mode

With interleaving, the maximum capture frequency doubles. However, the performance of interleave is limited by the stamper 1 to stamper 2 enable time of 6.25 ns and the event signal frequency.

In the example shown in the following figure, time stamper 2 is enabled immediately after time stamper 1 has stamped an event. Each stamper is no faster than in any other mode; however, each pair of events can be detected and stamped closer to each other. In doing this, you must consider the stamper-to-stamper offset error. To perform more accurate high frequency measurements, use precount between events mode when interleaving as shown in the figure.



Interleave Combined with Precount

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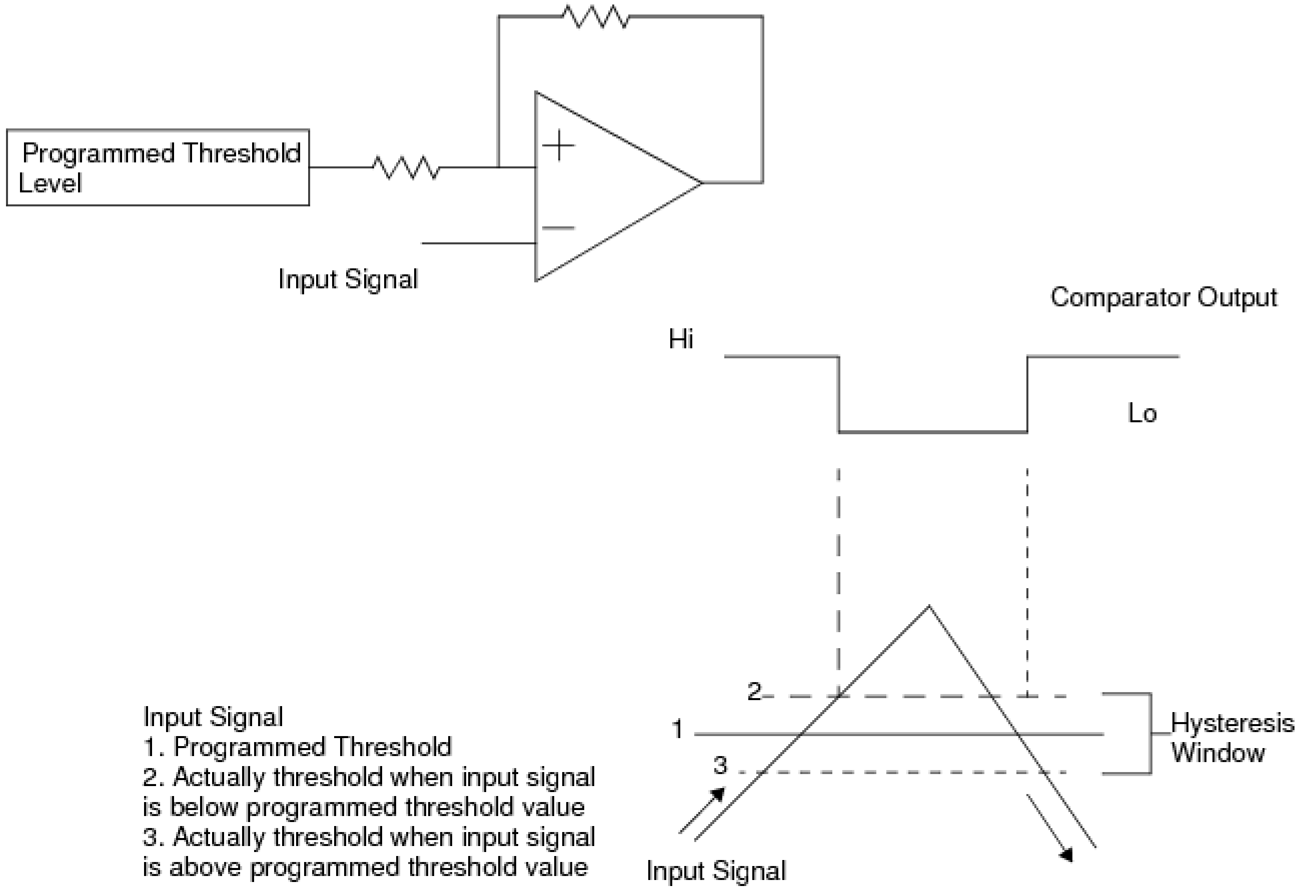
VHFAC TMU Theory of Operation : Hysteresis

Hysteresis

Comparator circuits chatter as the threshold is crossed if the input signal is moving slowly (where “slowly” is a relative term). When the input signal is far below the threshold (trip point of the comparator which causes the output to change state) of the comparator the output is at a steady state. As the input approaches the threshold level, it is moving slowly, and the comparator output may change states multiple times as the threshold is crossed. This only gets worse with the addition of noise on the input signal or in the comparator circuit itself.

Hysteresis is used to correct this undesired behavior. Hysteresis changes the threshold level as the input crosses it, in such a way that once the comparator changes state it remains in the new state. With the comparator in a new state, the input signal is farther away from the new threshold level. Since the input signal is farther away from the threshold after the state changes, the comparator output does not become confused and change state.

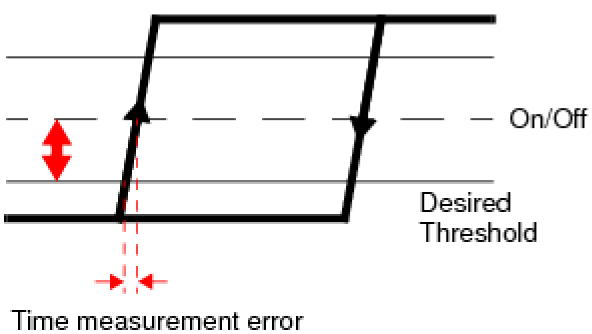
The input signal is made farther away by applying a small amount of positive feed back from the comparators output back into its threshold level input to get it to change slightly as the threshold is crossed. The actual threshold level is slightly different depending on if the input signal is below the threshold or above the threshold. The difference between the two levels is the threshold window.



Hysteresis

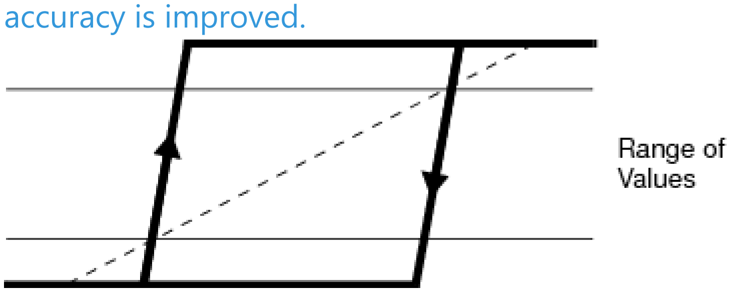
Programmable Hysteresis

The VHFAC TMU can provide programmable hysteresis. The hysteresis can be programmed to a value from 1 to 40 mV with a resolution of 0.5 mV.



Hysteresis Error Produces Time Measurement Error

With the TMU programmable hysteresis feature, stable measurements can be made and threshold accuracy is improved.



TMU Programmable Hysteresis Helps Make Stable Measurements

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VHFAC TMU Theory of Operation : VHFACTime Alarms

VHFACTime Alarms

An alarm indicates that the hardware is not operating as expected given the way the instrument has been programmed. It indicates that a condition exists that could potentially invalidate a measurement. The following is a list of supported alarms for VHFAC Time:

Measurement Incomplete	This alarm occurs if a measurement is stopped before the required number of samples has been captured. If a measurement is in progress and another Start microcode is executed, the current measurement stops and this alarm occurs.
DGS Alarm	Occurs when there is a difference in voltage between the DGS wire connected to the channel and the DIB ground.
StartEventnotArrival	Occurs if a measurement does not complete within the programmed time range (either the 10 ms or 20 s time range).
ADC Out Of Range	This alarm occurs if a Time stamper ADC measures voltage out of range. If the TMU calibration successfully completes, this alarm will not occur during normal measurement.

See also VHFAC Time Alarm in the debug display section.

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